

ELEC4631 Tutorial 5

1. Let S be a second order LTI state-space system with state $x = (x_1, x_2)^T$ given by:

$$S = \left(\begin{bmatrix} 5 & -14 \\ 2 & -6 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 & -5 \end{bmatrix}, 2 \right).$$

Perform a state transformation to a new state $z = (z_1, z_2)^T$ with $z_1 = 5x_1 - 2x_2$ and $z_2 = -2x_1 + x_2$. Determine the transformed state-space equations in the state z .

2. Consider again the system S from Problem 1. Answer the following questions:
- (a) Transform system into control canonical form.
 - (b) Transform system into modal canonical form.
 - (c) Transform system into observer canonical form.
 - (d) Compute the system transfer function.
 - (e) Compute a linear state feedback control law $u(t) = -Kx(t)$ such that $A - BK$ has a pair of conjugate eigenvalues at $-1 \pm j$.

3. Show that the two state space systems,

$$S_1 = \left(\begin{bmatrix} -1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \end{bmatrix}, 1 \right), S_2 = \left(\begin{bmatrix} -1 & 0 \\ 0.1 & -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \end{bmatrix}, 1 \right),$$

have the same transfer function but that they are not equivalent. That is, there does not exist an invertible matrix T that transforms S_1 into S_2 . (**Hint:** $\det(AB) = \det(A)\det(B)$).

4. Show that the matrix T that transforms a controllable system $S = (A, B, C, D)$ into control canonical form can be written as

$$T = \begin{bmatrix} t_1 \\ t_1 A \\ \vdots \\ t_1 A^{n-1} \end{bmatrix},$$

where t_1 denotes the last row of the inverse of the controllability matrix.